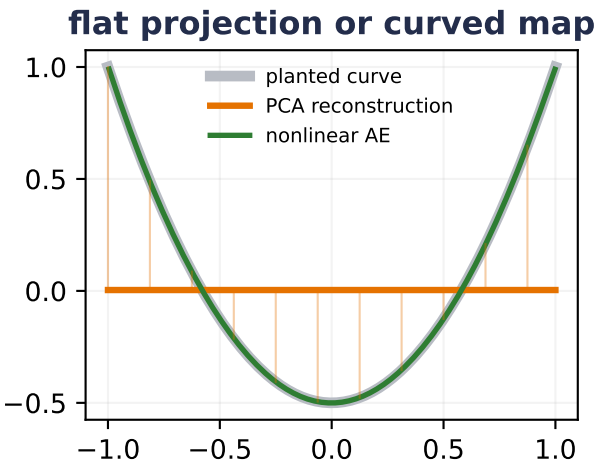
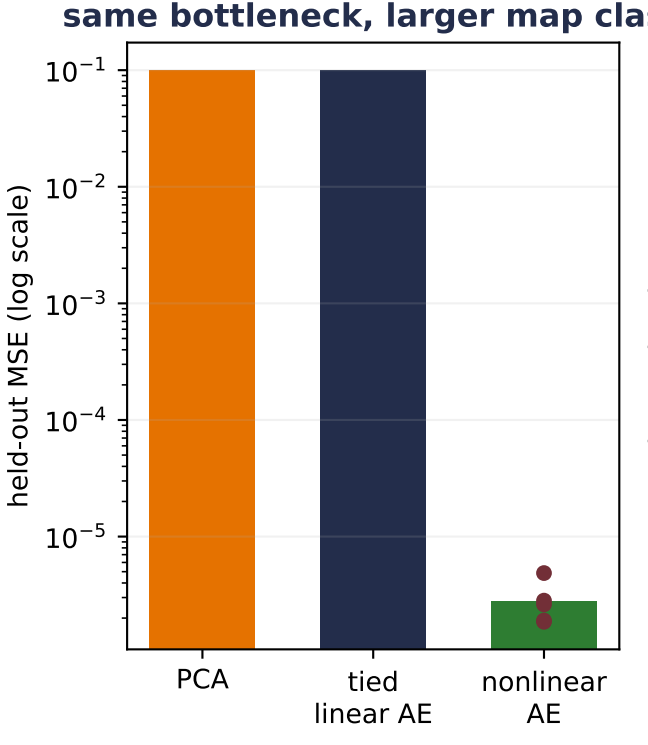
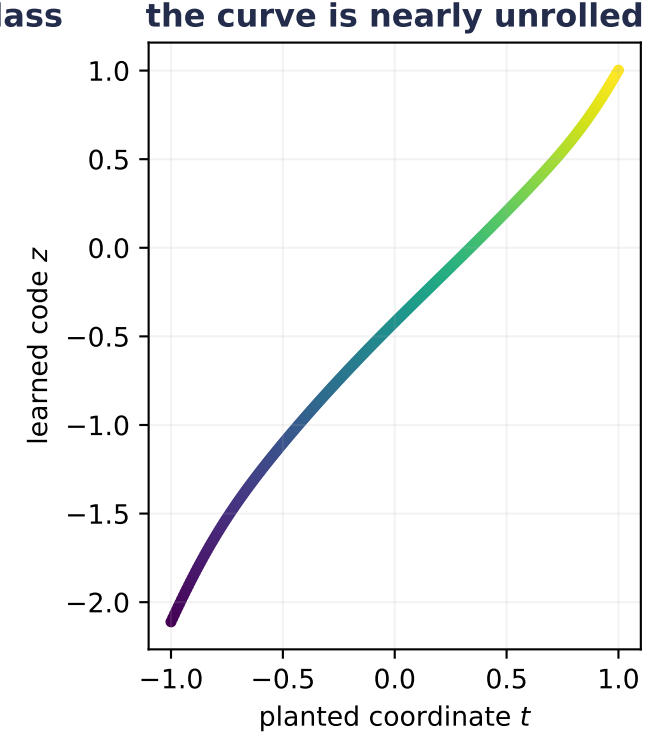

flat projection or curved map
planted curve
PCA reconstruction
nonlinear AE
1.0
0.5
0.0
−0.5
−1.0
−0.5
0.0
0.5
1.0


same bottleneck, larger map class
held-out MSE (log scale)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
PCA
tied
linear AE
nonlinear
AE


the curve is nearly unrolled
learned code z
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1.0
−0.5
0.0
0.5
1.0
planted coordinate t